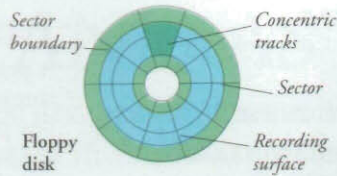


Computer disks

Programs and data can be stored on computer disks. Magnetic disks record data as magnetic patterns in tiny iron particles that coat the disks' surfaces. A hard disk is a stack of magnetic disks inside a computer. Other types of disk include compact discs (CDs) and magneto-optical disks. A device called a disk drive is used to store data on disks and retrieve it.



How a disk drive works

Disks arrange data into divisions called tracks and sectors. A disk drive has a magnetic read-write head that "reads" data from, or "writes" data to, a specific sector and track on the disk. In CD and optical disk drives, the read-write head is a laser beam.

Types of computer disk



Operating system allows this girl to use a program for learning Spanish.



Operating system

Every computer has a program called an operating system (OS) that controls its basic functions. The OS is always at work "behind the scenes" when other applications are running. A graphical user interface (GUI) often forms part of a computer's OS. The GUI allows the computer operator to use a mouse to move information or to run programs.

Steve Jobs and Steve Wozniak

The Apple Computer company – one of the world's largest – was founded in 1976 by Steve Jobs (b. 1955) and Steve Wozniak (b. 1950), who wanted to make computers affordable to ordinary people. Their 1977 Apple II computer was the first PC made for the mass market. It was hugely successful, because users of the Apple II needed no prior knowledge of electronics or computing.



Types of computer

Most schools and many homes have personal computers, but there are many other types of computer, both larger and smaller than a PC. Some computers enable people to work while they are travelling; others are designed purely for entertainment. Large, powerful, high-speed computers are often used to process information for many people at once, or perform many tasks simultaneously.



Dedicated computer

While some computers can carry out many different tasks, others are "dedicated", meaning that they are designed for one specific purpose. A familiar example of a dedicated computer system is a games computer.



Laptop

Many business people take small, portable PCs called laptops with them when they travel. A laptop contains rechargeable batteries that allow it to function on trains, buses, outdoors – in fact, almost anywhere.



Mainframe

Most large organizations have a mainframe computer that can be used by many people at once, each working at a separate monitor, or terminal. A mainframe must be kept in a cold, dry environment.



Supercomputer

The most powerful computers are supercomputers. They are used to perform very complex tasks, such as forecasting future weather systems or analysing gravity and black holes in space. The first supercomputer was launched in 1976 by Cray.

Timeline

1642 Blaise Pascal invents the first mechanical adding machine.

1834 Charles Babbage designs a mechanical computer, which he calls the Difference Engine.

1939 John Atanasoff, an American, completes the first electronic computer.



Commodore personal computer, 1970

1945 ENIAC, the world's first automatic computer, is built in the USA.

1967 Keyboards are used for data entry.

1970 Floppy disks appear.

Late 1970s Mass-produced PCs are introduced.

1975 The first portable computer is introduced.

Late 1970s The Xerox Corporation invents the graphical user interface.

1983 The mouse is first used on an Apple computer.

1985 Computer CDs appear.

Assorted software programs



1980s Sales of PCs soar.

1990 IBM Pentium PC performs 112 million instructions per second.

2000 Mobile phones and other hand-held devices are designed to include computing facilities, software, and Internet.

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